Artificial Neurons

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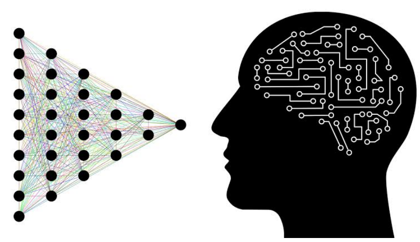
1 Understanding Neurons

1.1 What is a neuron

Biological neurons are cells in your brain whose primary purposes are to collect, process and transmit information throughout your body with electrochemical signals. They act like post offices, constantly receiving and sending messages. Our neurons take a much deeper place in our daily life when we understand that they are the ones responsible for decision making and the management of our emotions. From moving ourselves, to remembering and feeling, the uses of our neurons are numerous.



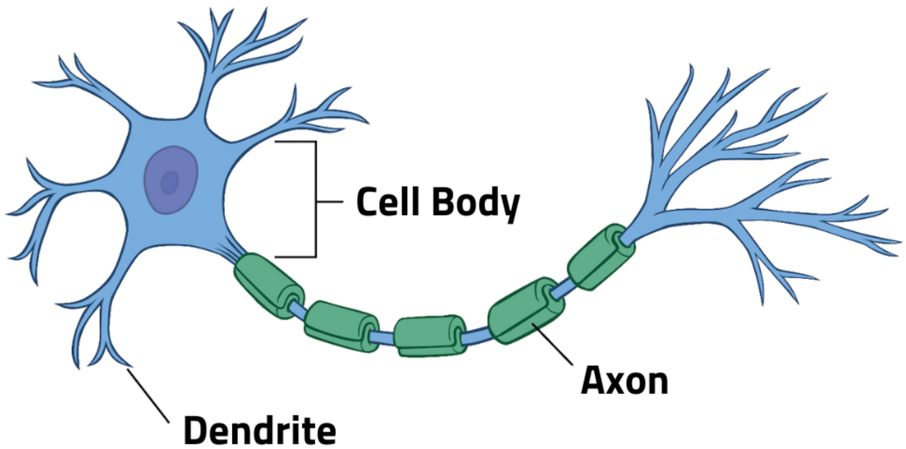
1.2 How do neurons function?

1.2.1 The parts of neurons

Neurons have three main parts:

- **The dendrite**: It receives signals from other neurons.
- **The soma** (*the cell's body*): The "processor" that sums up the incoming signals.
- **The Axon**: The output of the signal that carries it away to the next neuron.

1.2.2 Representation of a neuron



1.2.3 The process of information in neurons

- **Chemical Input**: The dendrites of the neurons receive a chemical signal(neurotransmitters).
- **Electrical Voltage**: The chemical signal received changes the charge of the cell and when it reaches a certain threshold, the neuron fires and electrical impulse — an action potential.
- **All-or-Nothing Output**: The firing of the impulse is binary and a dichotomous — either the neuron fires or it doesn't but it can't fire halfway.
- **The Synaptic Gap**: When the impulse reaches the end of the axon, it triggers the shipment of other neurotransmitters to the next neuron through a tiny gap(*The synapse*).

2 Learning more about Artificial Neurons

2.1 What is an artificial neuron ?

An artificial neuron (often called a perceptron) is a piece of mathematical code used in **ANNs** also known as Artificial Neural Networks — the technology behind machine learning and Artificial Intelligence (AI) — that mimics the behavior of a biological neuron but translates it to math.

2.2 How does it work ?

The process of information in artificial neurons consists in three steps:

- **The input**: The data (x) is fed to the neuron
- **The introduction of the weight and the bias**: the data is multiplied by it's weight (w)— It's importance — and the product is summed up with the bias that adjusts how easy it is for the neuron to fire ; it can be either positive or negative
- **Activation function**: The weighted inputs are added together using the following formula: z = $\sum_{i=1}^{n} w_i x_i + b$. The number resulting is then passed through an activation function that give the output(the most know being ReLU and Sigmoid)

2.3 It's applications

Nowadays, Artificial Neurons are used in many fields, form biology — by predicting how proteins fold — to face recognition — one example being the Face-id —, predicting analytics and many others.

2.4 Why is it revolutionary ?

The use of artificial neurons become more efficient than that of traditional step by step computers, because it can solve more complex problems by adapting it's self every time it makes a mistake(autodidact).

Taking the example of one task, recognizing the picture of a cat, a step by step computer would use many conditional blocs for instance in this case, "if it has to triangles(for the ears) and two round circles(the eyes) under the triangle....." but what if the picture is the back of a cat, the computer wouldn't be able to recognize it, just because the position of the cat has changed.

The artificial neuron on the other hand by giving him the picture of one millions of cats would adjust the weight of each information every time and the more pictures of cats analyzed, the less mistakes there would be and that is the very foundation of **Machine Learning**.

3 The link between artificial neurons and biological neurons

3.1 The historical link

The most direct link is that biological neurons served as the conceptual blueprint for artificial ones.

In 1943, a neuro-physiologist (Warren McCullough) and a mathematician (Walter Pitts) teamed up. They wanted to understand how the human brain thinks. They wrote a paper trying to turn the basic biological process of a neuron into a mathematical formula.

They took three major concepts from biological neurons:

- **The input**: Just like a biological neuron receives many inputs, so does the artificial neuron(x_1, x_2, x_3....)
- **The threshold**: The biological neuron requires reaching a certain threshold of chemical input to fire and the the artificial neuron in a way does the same, by reaching a certain mathematical threshold before firing
- **The Network**: Warren McCullough and Walter Pitts realized that a single neuron couldn't do much but when connecting it with other artificial neurons, it becomes even more efficient, this is how was created the first "Artificial Brain" also known as the Artificial Neural Network (ANN)

3.2 The learning Link

In biology, there is a famous rule called Hebb's Law (discovered by Donald Hebb in 1949). It explains how humans learn: when two biological neurons repeatedly fire at the same time, the chemical connection (the synapse) between them strengthens. If you practice the piano, those specific neural pathways get stronger.

In a way, the artificial neuron does the same by introducing weight to the inputs. When AI is training and gets a correct answer, the weight between the artificial neurons responsible increase but when it makes a mistake, it weakens.

4 Conclusion

Artificial neurons are a new way of how machines can work efficiently and ironically, that way is by modeling them self with the human's brain their own creator. We can find many advantages to artificial neurons but given the actual debates about digitization, ***can we really say it is only a benefit for the progress of humanity ?***